

Kimia Farma Business Performance Analysis (2020 - 2023)

Kimia Farma - Big Data Analytics

Presented by
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Big Data Analyst Virtual Intern at Kimia Farma

A recent **Agribusiness graduate** and **aspiring Data Analyst** with cross disciplinary experience in creative media, public relations, and student organizations, complemented by hands on operational exposure. Developing a strong analytical profile through **advanced Excel** and **SQL** to generate business insights and support data driven decision.



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Education

UIN Syarif Hidayatullah Jakarta | Program Studi Agribisnis | GPA: 3,69

Sept, 2021 - Okt 2025

Universitas Padjadjaran | MK Agribisnis Internasional | Grade: A

Mar, 2024 - Jul, 2024

Organization

HMJ Agribisnis UIN Jakarta | Staff Media Informasi

Apr, 2022 - Dec, 2023

Courses and Certification

Microsoft Excel Advanced | MySkill

Jan 2026

Intro to Data Analytics | RevoU

Oct 2025

About Company



PT Kimia Farma (Persero) Tbk is a pioneer in the Indonesian pharmaceutical industry, having developed into a company providing integrated healthcare services from upstream to downstream. Kimia Farma's core business areas include: Pharmaceutical Manufacturing supported by Research and Development; Distribution and Trading; Marketing; Pharmaceutical Retail; Clinical Laboratories and Health Clinics.

Vision:

To become the premier integrated Healthcare company of choice and generate sustainable value.

Mission:

- 1. To conduct business activities in the chemical and pharmaceutical industries, trade and distribution networks, pharmaceutical retail and healthcare services, and asset optimization.
- 2. To manage the company with good corporate governance and operational excellence, supported by professional human resources.
- 3. To provide added value and benefits to all stakeholders.

Core Value:

The company has established **AKHLAK** as the Kimia Farma Group's core values, replacing ICARE. The acronym for **AKHLAK** stands for **Amanah (Trustworthy)**, **Kompeten (Competent)**, **Harmonis (Harmonious)**, **Loyal**, **Adaptif (Adaptive)**, and **Kolaboratif (Collaborative)**. It serves as the identity and unifying force of the work culture, supporting sustainable performance improvement in every state-owned enterprise.

About the **Project**

Problem Statement

This project was driven by the company's need to evaluate business performance for the period 2020 to 2023. Currently, the primary challenge is integrating disparate data into a single, centralized source of information in BigQuery. This integration is essential for management to more accurately identify anomalies in branch performance.

Objective

As a Big Data Analyst at PT Kimia Farma, this project aims to:

- **Evaluate** Kimia Farma's business performance in-depth and comprehensively for the 2020 to 2023 period.
- **Integrate** disparate data into a single, centralized dataset.
- **Identify** branch performance, particularly the ratings for each branch

[Project explanation video here!](#)

About the **Tools**



Data File Format
.csv



Google BigQuery SQL



Google Data Studio
(Looker)

About The Dataset

There are 4 types of data that will be input into the dataset and used for analysis and dashboard creation.

All of it uploaded to the dataset in BigQuery from .csv format

☐	Field name	Type	Mode
☐	Inventory_ID	STRING	NULLABLE
☐	branch_id	INTEGER	NULLABLE
☐	product_id	STRING	NULLABLE
☐	product_name	STRING	NULLABLE
☐	opname_stock	INTEGER	NULLABLE

kf_inventory

☐	Field name	Type	Mode
☐	branch_id	INTEGER	NULLABLE
☐	branch_category	STRING	NULLABLE
☐	branch_name	STRING	NULLABLE
☐	kota	STRING	NULLABLE
☐	provinsi	STRING	NULLABLE
☐	rating	FLOAT	NULLABLE

kf_kantor_cabang

☐	Field name	Type	Mode
☐	transaction_id	STRING	NULLABLE
☐	date	DATE	NULLABLE
☐	branch_id	INTEGER	NULLABLE
☐	customer_name	STRING	NULLABLE
☐	product_id	STRING	NULLABLE
☐	price	INTEGER	NULLABLE
☐	discount_percentage	FLOAT	NULLABLE
☐	rating	FLOAT	NULLABLE

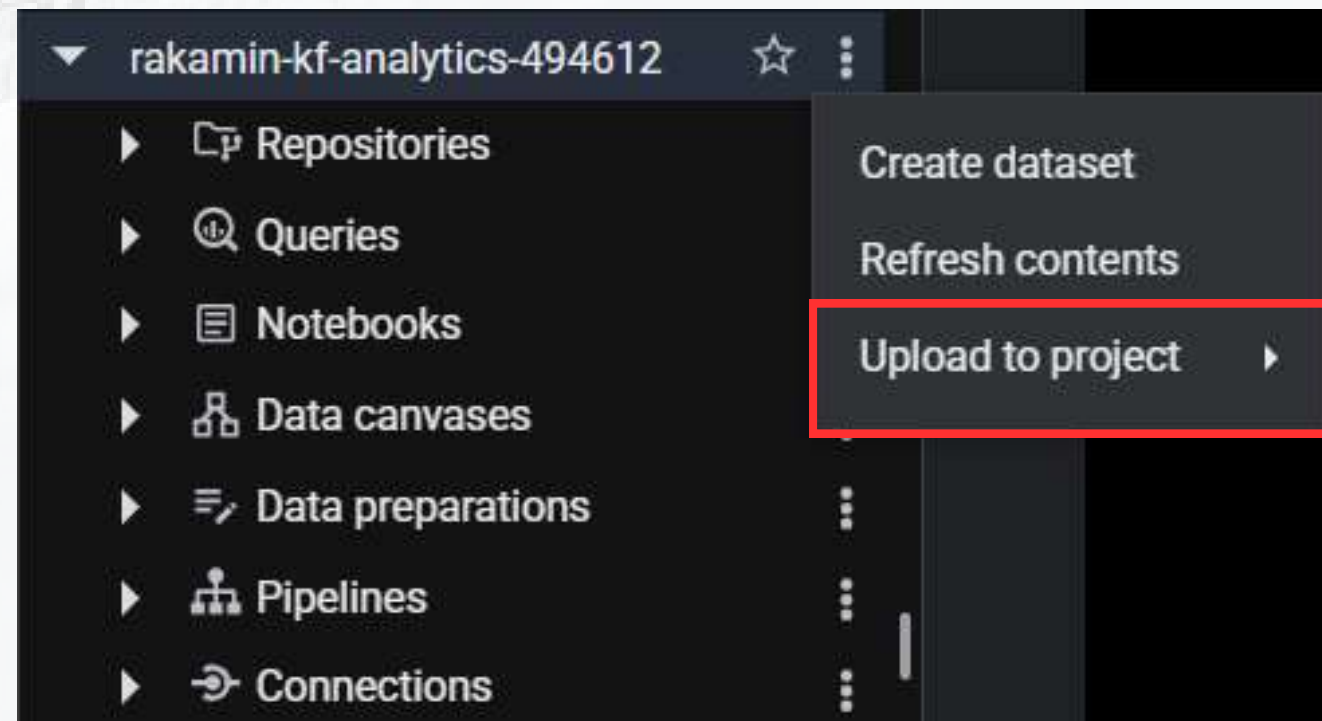
kf_final_transaction

☐	Field name	Type	Mode
☐	product_id	STRING	NULLABLE
☐	product_name	STRING	NULLABLE
☐	product_category	STRING	NULLABLE
☐	price	INTEGER	NULLABLE

kf_product

The Process

1. Importing the dataset into BigQuery



Import 4 raw datasets (final_transaction, inventory, kantor_cabang, and product) in .csv format into BigQuery to one dataset named kimia_farma.

The Process

2. Data Integration & Transformation

```
CREATE OR REPLACE TABLE `kimia_farma.analysis_table` AS
WITH
base_data AS (
  SELECT
    tran.transaction_id,
    tran.date,
    tran.branch_id,
    cab.branch_name,
    cab.kota,
    cab.provinsi,
    cab.rating AS rating_cabang,
    tran.customer_name,
    tran.product_id,
    prod.product_name,
    prod.price AS actual_price,
    tran.discount_percentage,
    tran.rating AS rating_transaksi,
    inv.opname_stock AS stok_tersedia
  FROM `kimia_farma.kf_final_transaction` AS tran
  LEFT JOIN `kimia_farma.kf_kantor_cabang` AS cab ON tran.branch_id = cab.branch_id
  LEFT JOIN `kimia_farma.kf_product` AS prod ON tran.product_id = prod.product_id
  LEFT JOIN `kimia_farma.kf_inventory` AS inv ON tran.branch_id = inv.branch_id AND tran.product_id = inv.product_id
),
```

*Simultaneously, rename all tables to:
tran for the transaction table,
cab for the branch table,
prod for the product table,
and inv for the inventory table.

Combine four raw data sets from the dataset into one new table for the final analysis and reporting to BI tools by combining all tables (transactions, inventory, branch offices, and products) using common table expression (CTE) and left join to combine all tables with the tran table as the fact table.

The Process

3. Gross Profit Percentage Calculation

```
profit_calculation AS (  
  SELECT *,  
    CASE  
      WHEN actual_price <= 50000 THEN 0.10  
      WHEN actual_price > 50000 AND actual_price <= 100000 THEN 0.15  
      WHEN actual_price > 100000 AND actual_price <= 300000 THEN 0.20  
      WHEN actual_price > 300000 AND actual_price <= 500000 THEN 0.25  
      ELSE 0.30  
    END AS percentase_gross_laba  
  FROM base_data  
) ,
```

Next, using the CASE WHEN function I calculate the gross profit percentage of each product automatically by classifying each product into margin categories ranging from 10%, 15%, 20%, 25%, to 30% based on its selling price (actual_price). This process allows for standardization and identification of product groups that contribute the largest margins.

The Process

4. Financial Metrics Calculation

```
final_metrics AS (  
  SELECT *,  
    -- Menghitung harga setelah diskon yang dibayar konsumen  
    (actual_price * (1 - discount_percentage)) AS nett_sales,  
    -- Menghitung laba bersih dari nett sales  
    ((actual_price * (1 - discount_percentage)) * persentase_gross_laba) AS nett_profit  
  FROM profit_calculation  
)
```

Next, calculate the key financial metrics which are **net sales** and **net profit**.

Net sales are calculated to determine the value of **revenue after** deducting **discounts** given to consumers, which are then multiplied by the previously determined gross profit percentage to obtain **net profit**.

The Process

5. Final Data Selection

```
SELECT
transaction_id,
date,
branch_id,
branch_name,
kota,
provinsi,
rating_cabang,
customer_name,
product_id,
product_name,
actual_price,
discount_percentage,
persentase_gross_laba,
nett_sales,
nett_profit,
rating_transaksi,
stok_terseedia,
FROM final_metrics;
```

Using SELECT statement on all relevant final columns to be used as dashboard data sources, starting from transaction identities, branch details, to previously calculated financial calculations.

The selection of the specific table from the final_metrics CTE needs to be paid attention so that all calculations and CTE columns that have been created can be processed when using the SELECT statement.

[For full SQL syntax click here!](#)

The Dashboard

[Link to the dashboard!](#)

Kimia Farma Performance Dashboard (2020-2023)

Pilih rentang tanggal

Provinsi

Nama Cabang

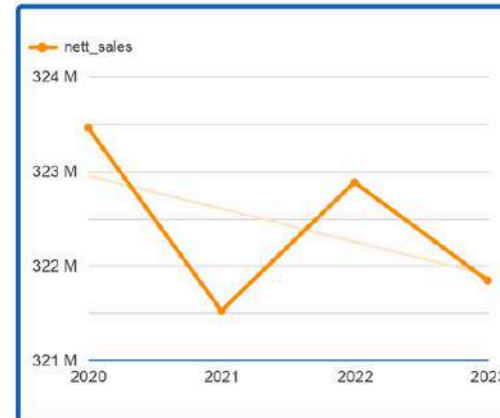
Total Pendapatan
1,29 T

Total Keuntungan
366,32 M

Total Transaksi
2,70 jt

Rata-rata Rating Transaksi
4,00

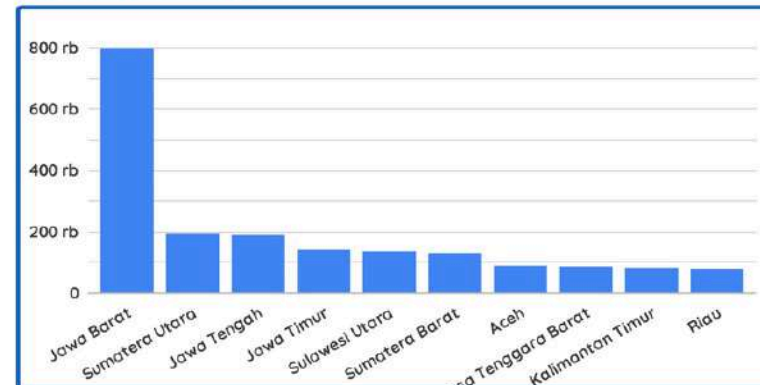
Sales Over Time (Pertahun)



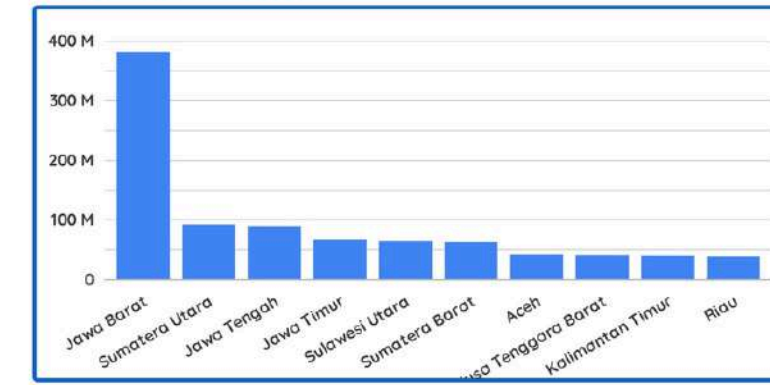
Total Keuntungan per Provinsi



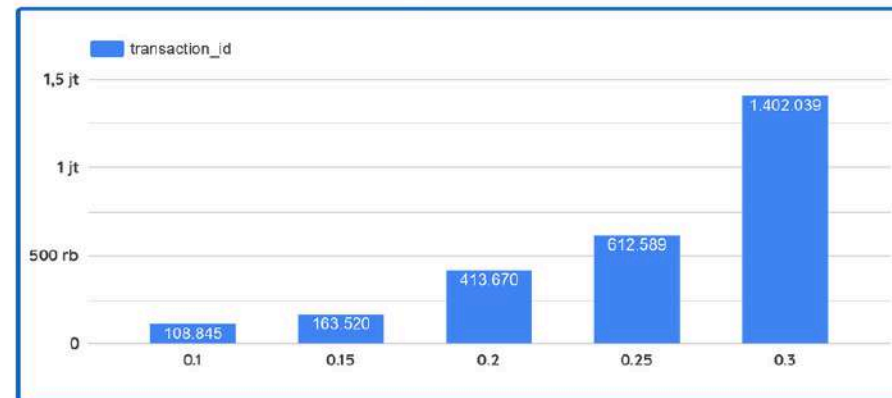
Top 10 Transaksi Provinsi



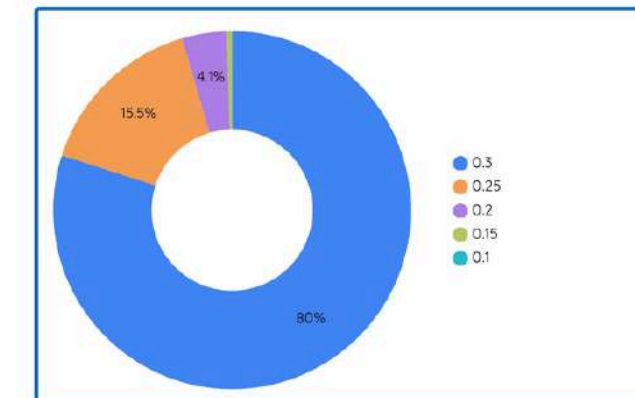
Top 10 Nett Sales Provinsi



Jumlah Transaksi per Persentase Gross Laba



Kontribusi Profit per Tier Laba



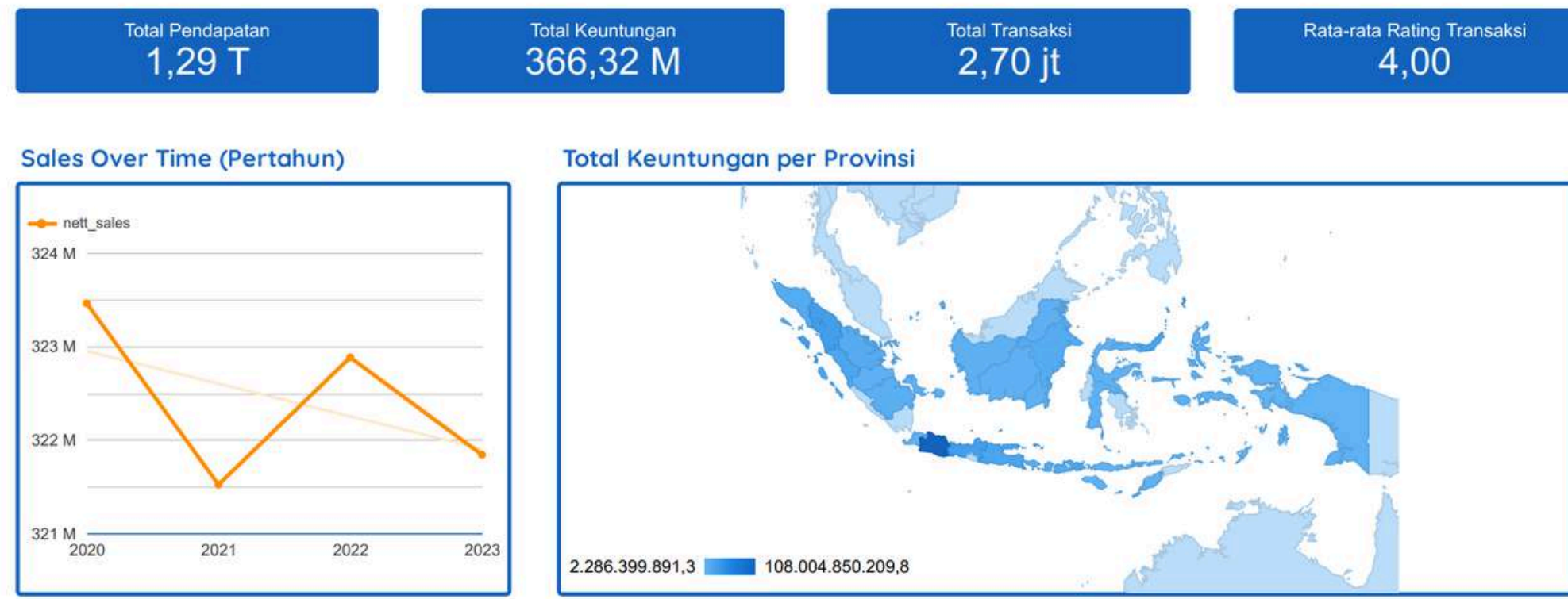
Top 5 Selisih Rating Cabang dengan Rating Transaksi

Cabang	Kota	Rating Caba...	Rating Transa...	Selisih Rati...
1. Kimia Farma - Klinik-Apotek-Laboratorium	Pematangsiantar	4,84	3,99	0,85
2. Kimia Farma - Klinik & Apotek	Jambi	4,78	4,01	0,77
3. Kimia Farma - Apotek	Sorong	4,73	3,99	0,74
4. Kimia Farma - Klinik-Apotek-Laboratorium	Batam	4,72	4,00	0,72
5. Kimia Farma - Klinik-Apotek-Laboratorium	Cilecap	4,72	4,00	0,72

Key Takeaways:

- Trend sales tahun 2020-2023 fluktuatif menurun
- Provinsi Jawa Barat menjadi revenue leader dengan sales sebesar 380 M
- Produk dengan laba 30% berkontribusi 80% terhadap total profit
- Produk dengan laba 30% juga menjadi produk dengan transaksi tertinggi sebanyak 348.915 kali
- Terdapat 5 cabang dengan selisih rating yang cukup besar menandakan adanya gap pada kualitas pelayanan terhadap customer

Insights from Dashboard



Key Takeaways:

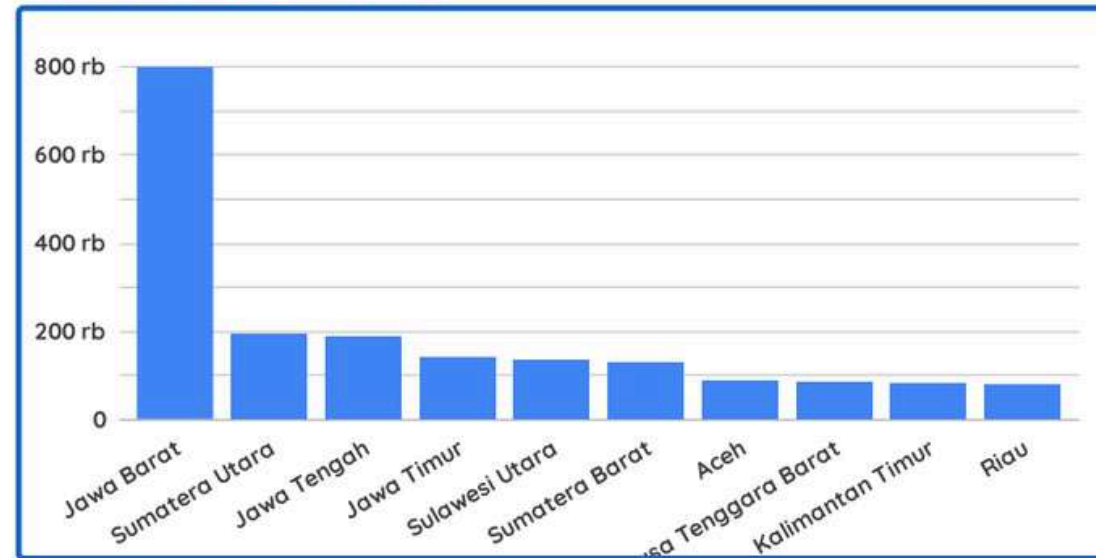
- During the 2020-2023 period, Kimia Farma achieved revenue of IDR 1.29 trillion with a net profit of IDR 366.32 billion from 2.7 million transactions, with a customer satisfaction rating of 4.
- Annual sales fluctuated, with the lowest decline in 2020-2021 reaching IDR 2 billion.
- West Java Province became the market leader.

Actionable Insight:

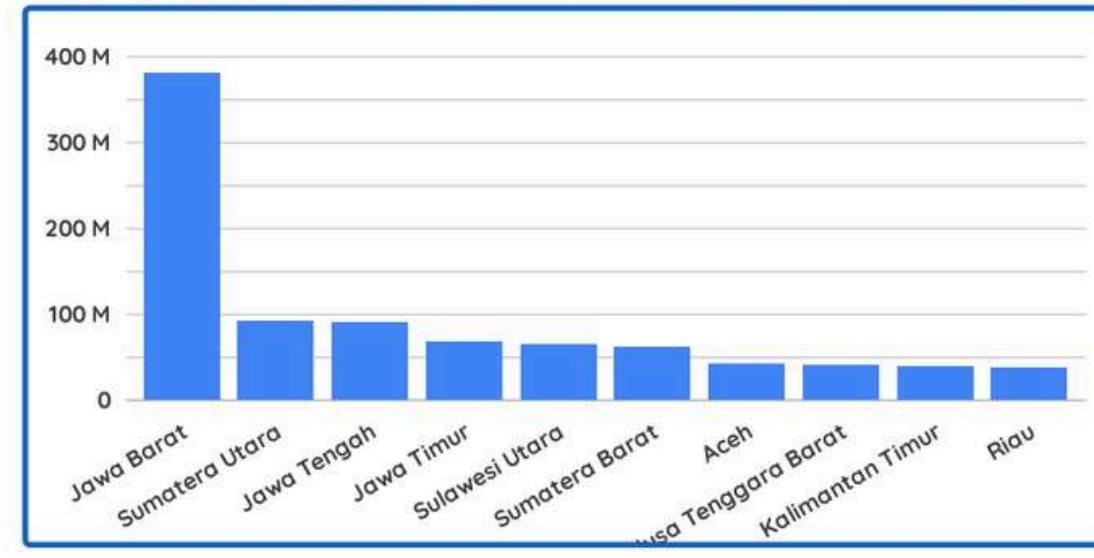
- Focus on maintaining and strengthening dominance in Java, particularly West Java, by optimizing and streamlining the supply chain and implementing customer loyalty programs to secure revenue streams.
- Analysis of external factors related to annual sales fluctuations is necessary to prepare mitigation strategies for the coming year.

Insights from Dashboard

Top 10 Transaksi Provinsi



Top 10 Nett Sales Provinsi



Key Takeaways:

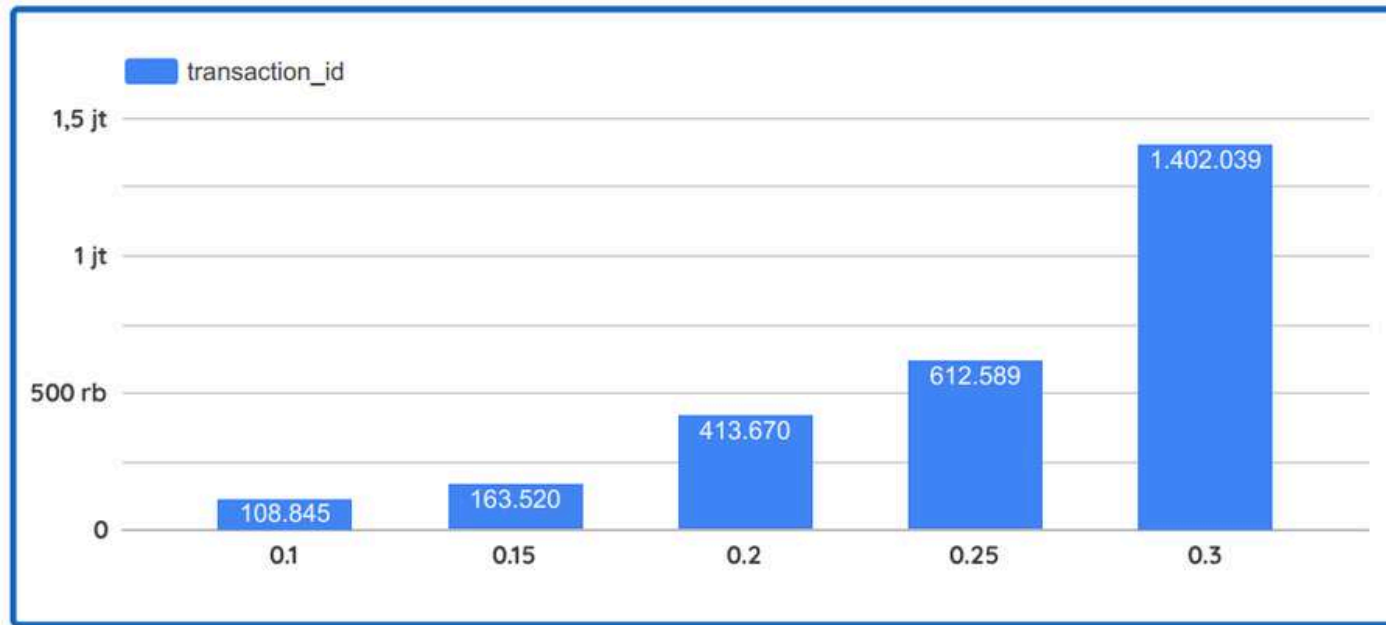
- Transactions in West Java reached over 800,000 and net sales approached IDR 400 billion. Kimia Farma is highly dependent on its performance in West Java.
- Average sales in the other top 10 provinces tended to be similar, meaning they were highly dependent on the number of customers visiting.

Actionable Insight:

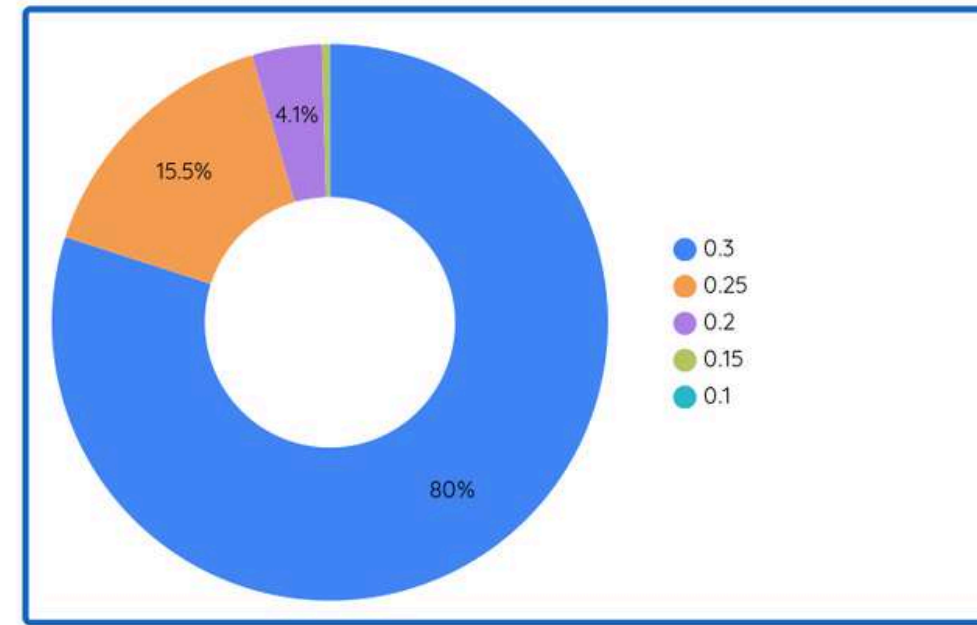
- Retaining customers in West Java Province through a customer loyalty program.
- Market penetration strategies in other provinces through promotions or collaborations with local healthcare facilities.
- Up-selling strategies to increase net sales without opening more branches by increasing customer spending.

Insights from Dashboard

Jumlah Transaksi per Persentase Gross Laba



Kontribusi Profit per Tier Laba



Key Takeaways:

- Products with a 30% margin contribute 80% to the company's profit with a total of 1.4 million transactions, the dependence of profits on products with a 30% margin is very high.

Actionable Insight:

- Prioritize stocking of products with a 30% margin across all branches to avoid losses due to stockouts.
- Up-selling by offering vitamins/supplements with a 30% margin when consumers purchase lower-tier medications.
- Reviewing the storage and logistics of products with 10% and 15% margins to avoid costs. It's better to reduce the variety or adjust the prices to increase margins.

Insights from Dashboard

Top 5 Selisih Rating Cabang dengan Rating Transaksi

	Cabang	Kota	Rating Caba...	Rating Transa...	Selisih Rati... ▾
1.	Kimia Farma - Klinik-Apotek-Laboratorium	Pematangsiantar	4,84	3,99	0,85
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5.	Kimia Farma - Klinik-Apotek-Laboratorium	Cilacap	4,72	4,00	0,72

Key Takeaways:

- There are 5 branches that have a difference between the physical rating of the branch and the rating of the customer transaction experience, indicating an imbalance between the physical quality of the branch and the customer shopping experience.

Actionable Insight:

- This requires a service audit and SOP check at problematic branches, especially those five.
- Evaluate stock and product availability at these branches, to mitigate low transaction ratings due to unavailable items, and provide retraining on customer excellence for staff at these branches.

The Closing Summary

Summary:

- Overall, the analysis for the 2020-2023 period shows that Kimia Farma has strong financial performance with total revenue of IDR 1.29 trillion. However, it is highly dependent on profits of products with 30% margin, which contribute 80% of the company's total profit.
- Although West Java is the largest national sales contributor, there are still branches in other regions whose transaction service quality cannot match the superior physical facilities of the building.
- Kimia Farma need to prioritize stock management on high-margin products, conduct service audits on anomalous branches, and begin more equitable market expansion outside Java.

As a closing, through this dashboard Kimia Farma now has strategic guidance to maintain its 80% profit margin and improve service in branches with high rating gaps. I believe this data is key to more stable growth in the years to come.

Thank You!

**I am very open to further discussion regarding this data insight
or other collaboration opportunities.**

Reach me out!

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 <https://github.com/ReghanMarciano>

